KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3131 | 3 Credits: 4 | Program Core | Source-grounded

Computer Organisation

● Visualize the basic building blocks of a computer.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3131 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3131.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering
Course code	3131
Course title	Computer Organisation
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

22 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Visualize the basic building blocks of a computer.
- ● Understand the memory hierarchy and technology, different ways of communicating
- with the I/O and interfaces.
- ● Familiarise the functional units and working of a processor unit.
- ● Provide knowledge on latest processor technologies.

Course outcomes

CO	Official outcome	Study level
CO1	16 Understanding memory of a computer system Explain the Input / Output organisation of a	See official syllabus
CO2	15 Understanding Computer System Illustrate the working of Processing Unit of a	See official syllabus
CO3	14 Understanding Computer System	See official syllabus
CO4	Illustrate the working of Microprocessors 12 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	system.	9	As specified
Module 2	Explain the Input / Output organisation of a Computer System	5	As specified
Module 3	Illustrate the working of Processing Unit of a Computer System	4	As specified
Module 4	Illustrate the working of Microprocessors	4	As specified

MODULE 1

system.

This module is presented from the official syllabus scope for Computer Organisation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: system.
- Official duration: As specified
- Official topic entries captured: 9

- The full source excerpt is preserved below for coverage checking.

1 Understanding Computer system. Illustrate the basic operational concepts of a

1 Understanding Computer system. Illustrate the basic operational concepts of a is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Computer system. Illustrate the basic operational concepts of a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding computer system. illustrate the basic operational concepts of a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Computer system. Illustrate the basic operational concepts of a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding Computer system. Illustrate the basic operational concepts of a [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding computer system.

1 Understanding computer system. is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding computer system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding computer system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding computer system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding computer system. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the bus structure 1 Understanding Explain the connection of memory to the

Explain the bus structure 1 Understanding Explain the connection of memory to the is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the bus structure 1 Understanding Explain the connection of memory to the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the bus structure 1 understanding explain the connection of memory to the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the bus structure 1 Understanding Explain the connection of memory to the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the bus structure 1 Understanding Explain the connection of memory to the □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding processor. Summarize the features of various

1 Understanding processor. Summarize the features of various is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding processor. Summarize the features of various using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding processor. summarize the features of various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding processor. Summarize the features of various means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 1 Understanding processor. Summarize the features of various □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Understanding semiconductor memories Outline the memory hierarchy with respect

3 Understanding semiconductor memories Outline the memory hierarchy with respect is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding semiconductor memories Outline the memory hierarchy with respect using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding semiconductor memories outline the memory hierarchy with respect should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding semiconductor memories Outline the memory hierarchy with respect means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding semiconductor memories Outline the memory hierarchy with respect □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding to speed, size and cost. Infer the effective memory access time of

2 Understanding to speed, size and cost. Infer the effective memory access time of is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding to speed, size and cost. Infer the effective memory access time of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding to speed, size and cost. infer the effective memory access time of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding to speed, size and cost. Infer the effective memory access time of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding to speed, size and cost. Infer the effective memory access time of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding cache memory

2 Understanding cache memory is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding cache memory using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding cache memory should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding cache memory means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding cache memory □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Explain the concept of virtual memory 2 Understanding Summarize the features of various secondary

Explain the concept of virtual memory 2 Understanding Summarize the features of various secondary is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of virtual memory 2 Understanding Summarize the features of various secondary using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of virtual memory 2 understanding summarize the features of various secondary should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of virtual memory 2 Understanding Summarize the features of various secondary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the concept of virtual memory 2 Understanding Summarize the features of various secondary [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding storage devices Computer System - Functional units - Basic operational concepts - Connection of the memory to the processor -Bus Structure -Semiconductor RAM Memories - Organisation of bit cells in a memory chip. Static memory (General features only) - Synchronous DRAM, Asynchronous DRAM, Double data rate SDRAM. Read Only Memory (General features only) - ROM, PROM, EPROM, EEPROM, Flash Memory. Memory Hierarchy (basic concept only) - Cache Memory (basic concept only) - Virtual Memory (basic concept only). Secondary Storage (General features only) - Magnetic Hard Disks - RAID - Optical Disks.

3 Understanding storage devices Computer System - Functional units - Basic operational concepts - Connection of the memory to the processor -Bus Structure -Semiconductor RAM Memories - Organisation of bit cells in a memory chip. Static memory (General features only) - Synchronous DRAM, Asynchronous DRAM, Double data rate SDRAM. Read Only Memory (General features only) - ROM, PROM, EPROM, EEPROM, Flash Memory. Memory Hierarchy (basic concept only) - Cache Memory (basic concept only) - Virtual Memory (basic concept only). Secondary Storage (General features only) - Magnetic Hard Disks - RAID - Optical Disks. is an official topic in Module 1 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding storage devices Computer System - Functional units - Basic operational concepts - Connection of the memory to the processor - Bus Structure -Semiconductor RAM Memories - Organisation of bit cells in a memory chip. Static memory (General features only) - Synchronous DRAM, Asynchronous DRAM, Double data rate SDRAM. Read Only Memory (General features only) - ROM, PROM, EPROM, EEPROM, Flash Memory. Memory Hierarchy (basic concept only) - Cache Memory (basic concept only) - Virtual Memory (basic concept only). Secondary Storage (General features only) - Magnetic Hard Disks - RAID - Optical Disks. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding storage devices computer system - functional units - basic operational concepts - connection of the memory to the processor -bus structure -semiconductor ram memories - organisation of bit cells in a memory chip. static memory (general features only) - synchronous dram, asynchronous dram, double data rate sdram. read only memory (general features only) - rom, prom, eprom, eeprom, flash memory. memory hierarchy (basic concept only) - cache memory (basic concept only) - virtual memory (basic concept only). secondary storage (general features only) - magnetic hard disks - raid - optical disks. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding storage devices Computer System - Functional units - Basic operational concepts - Connection of the memory to the processor -Bus Structure -Semiconductor RAM Memories - Organisation of bit cells in a memory chip. Static memory (General features only) - Synchronous DRAM, Asynchronous DRAM, Double data rate SDRAM. Read Only Memory (General features only) - ROM, PROM, EPROM, EEPROM, Flash Memory. Memory Hierarchy (basic concept only) - Cache Memory (basic concept only) - Virtual Memory (basic concept only). Secondary Storage (General features only) - Magnetic Hard Disks - RAID - Optical Disks. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding storage devices Computer System - Functional units - Basic operational concepts - Connection of the memory to the processor -Bus Structure -Semiconductor RAM Memories - Organisation of bit cells in a memory chip. Static memory (General features only) - Synchronous DRAM, Asynchronous DRAM, Double data rate SDRAM. Read Only Memory (General features only) - ROM, PROM, EPROM, EEPROM, Flash Memory. Memory Hierarchy (basic concept only) - Cache Memory (basic concept only) - Virtual Memory (basic concept only). Secondary Storage (General features only) - Magnetic Hard Disks -

RAID - Optical Disks. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
system.
            Describe various functional units of a
  M1.01                                                        1
Understanding
            Computer system.
            Illustrate the basic operational concepts of a
  M1.02                                                        1
Understanding
            computer system.
  M1.03     Explain the bus structure                          1
Understanding
            Explain the connection of memory to the
  M1.04                                                        1
Understanding
            processor.
            Summarize the features of various
  M1.05                                                        3
Understanding
            semiconductor memories
            Outline the memory hierarchy with respect
  M1.06                                                        2
Understanding
            to speed, size and cost.
            Infer the effective memory access time of
  M1.07                                                        2
Understanding
            cache memory
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the Input / Output organisation of a Computer System

This module is presented from the official syllabus scope for Computer Organisation. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the Input / Output organisation of a Computer System
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in

3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in is an official topic in Module 2 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mapped i/o and program controlled i/o. illustrate the working of interrupts in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an

3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an is an official topic in Module 2 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding accessing i/o devices. demonstrate direct memory access as an should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding I/O mechanism for high speed devices.

3 Understanding I/O mechanism for high speed devices. is an official topic in Module 2 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding I/O mechanism for high speed devices. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding i/o mechanism for high speed devices. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding I/O mechanism for high speed devices. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding I/O mechanism for high speed devices. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer

Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer is an official topic in Module 2 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the features of standard interfaces 3 understanding summarize the features of computer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding peripherals Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts - Direct Memory Access (basic concept only) Standard I/O interfaces (General features only) - PCI, SCSI, USB Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse, Scanner - Output Devices - Video Displays, Flat Panel Displays, Printers, Graphics Accelerators.

3 Understanding peripherals Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts - Direct Memory Access (basic concept only) Standard I/O interfaces (General features only) - PCI, SCSI, USB Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse, Scanner - Output Devices - Video Displays, Flat Panel Displays, Printers, Graphics Accelerators. is an official topic in Module 2 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding peripherals Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts - Direct Memory Access (basic concept only) Standard I/O interfaces (General features only) - PCI, SCSI, USB Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse, Scanner - Output Devices - Video Displays, Flat Panel Displays, Printers, Graphics Accelerators. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding peripherals input/output organisation - accessing of i/o devices - memory mapped i/o, program controlled i/o, interrupts -

interrupt hardware, handling multiple interrupts - direct memory access (basic concept only) standard i/o interfaces (general features only) - pci, scsi, usb computer peripherals (general features only) - input devices - keyboard, mouse, scanner - output devices - video displays, flat panel displays, printers, graphics accelerators. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding peripherals Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts - Direct Memory Access (basic concept only) Standard I/O interfaces (General features only) - PCI, SCSI, USB Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse, Scanner - Output Devices - Video Displays, Flat Panel Displays, Printers, Graphics Accelerators. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding peripherals Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts - Direct Memory Access (basic concept only) Standard I/O interfaces (General features only) - PCI, SCSI, USB Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse, Scanner - Output Devices - Video Displays, Flat Panel Displays, Printers, Graphics Accelerators. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the Input / Output organisation of a Computer System
             Outline I/O interfacing with memory
  M2.01                                                          3
Understanding
             mapped I/O and Program controlled I/O.
             Illustrate the working of interrupts in
  M2.02                                                          3
Understanding
             accessing I/O devices.
             Demonstrate Direct Memory Access as an
  M2.03                                                          3
Understanding
             I/O mechanism for high speed devices.
  M2.04      Outline the features of Standard Interfaces         3
Understanding
             Summarize the features of computer
  M2.05                                                          3
Understanding
             peripherals
             Series Test – I                                    1.5
Contents:
Input/Output Organisation - Accessing of I/O devices - Memory mapped I/O, Program
controlled I/O, Interrupts - Interrupt hardware, Handling Multiple Interrupts -
Direct
Memory Access (basic concept only)
Standard I/O interfaces (General features only) - PCI, SCSI, USB
Computer Peripherals (General features only) - Input Devices - Keyboard, Mouse,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Illustrate the working of Processing Unit of a Computer System

This module is presented from the official syllabus scope for Computer Organisation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the working of Processing Unit of a Computer System
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding processor

2 Understanding processor is an official topic in Module 3 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding processor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding processor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding processor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding processor □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate

Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate is an official topic in Module 3 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the execution of an instruction. 6 understanding illustrate the various methods to generate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding control signals

4 Understanding control signals is an official topic in Module 3 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding control signals using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding control signals should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding control signals means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Understanding control signals [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining .

Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining . is an official topic in Module 3 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining . using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the principle of pipelining 2 understanding processing unit : fundamental concepts - register transfers, performing arithmetic and logic operations, fetching a word from memory, storing a word in memory - execution of a complete instruction, branch instructions - hardwired control - complete processor - microprogrammed control - pipelining . should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining . means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining . □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the working of Processing Unit of a Computer System

             Explain the internal functional units of a
  M3.01                                                        2
Understanding
             processor
  M3.02      Illustrate the Execution of an instruction.       6
Understanding
             Illustrate the various methods to generate
  M3.03                                                        4
Understanding
             control signals
  M3.04      Outline the principle of Pipelining               2
Understanding

Contents:
Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic
and
Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution
of a Complete Instruction, Branch Instructions - Hardwired Control - Complete
Processor -
Microprogrammed Control - Pipelining .
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate the working of Microprocessors

This module is presented from the official syllabus scope for Computer Organisation. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the working of Microprocessors
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088

Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088 is an official topic in Module 4 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the history of microprocessors 1 understanding illustrate the architecture 8086/8088 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Understanding Microprocessor.

5 Understanding Microprocessor. is an official topic in Module 4 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Microprocessor. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding microprocessor. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Microprocessor. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 5 Understanding Microprocessor. [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible] [illegible].

Outline architecture of a Pentium Processor

Outline architecture of a Pentium Processor is an official topic in Module 4 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline architecture of a Pentium Processor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline architecture of a pentium processor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline architecture of a Pentium Processor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Outline architecture of a Pentium Processor [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 - Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization

Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization is an official topic in Module 4 of Computer Organisation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the features of multicore processors 3 understanding basic x86 architecture - role of microprocessor in micro computer – brief history of microprocessors (with specific insight into x86 family - features of 8086 -internal block diagram of 8086 – execution unit – bus interface unit – general architecture of a pentium processor - multi core processor text / reference: author /book t/r title t1 carl hamachar, computer organization, mcgraw hill, 5th ed. lyla b das, the x86 microprocessors- architecture, programming and t2 interfacing, pearson william stallings, computer organization and architecture, pearson r1 education , 8th ed. r2 morris mano, computer system architecture , prentice hall of india- 2002 r3 john hayes, computer architecture and organization ,mcgraw hill-1998 sunil mathur, microprocessor 8086 architecture, programming and r4 interfacing, phi,2011 online resources sl.no website link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/computer-architecture-and-organization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings,

Aspect	What to prepare
	Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the working of Microprocessors
  M4.01       Summarize the history of Microprocessors          1
Understanding
              Illustrate the architecture 8086/8088
  M4.02                                                         5
Understanding
              Microprocessor.
  M4.03       Outline architecture of a Pentium Processor       3
Understanding
  M4.04       Outline the features of Multicore Processors      3
Understanding
               Series Test – II                                1.5

Contents:
Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history
of
Microprocessors (with specific insight into x86 family - Features of 8086 -
Internal Block
Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a
Pentium Processor - Multi core Processor

Text / Reference:


                                            Author /Book
  T/R
                                                Title
   T1       Carl Hamachar, Computer Organization, McGraw Hill, 5th ed.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding Computer system. Illustrate the basic operational concepts of a. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding Computer system. Illustrate the basic operational concepts of a*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding computer system.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding computer system.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the bus structure 1 Understanding Explain the connection of memory to the. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the bus structure 1 Understanding Explain the connection of memory to the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding processor. Summarize the features of various. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding processor. Summarize the features of various*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding accessing I/O devices. Demonstrate Direct Memory Access as an*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding I/O mechanism for high speed devices.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding I/O mechanism for high speed devices.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the features of Standard Interfaces 3 Understanding

Summarize the features of computer. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the features of Standard Interfaces 3 Understanding Summarize the features of computer*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding processor. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding processor*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the Execution of an instruction. 6 Understanding Illustrate the various methods to generate*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding control signals. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding control signals*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining .. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the principle of Pipelining 2 Understanding Processing Unit : Fundamental Concepts - Register Transfers, Performing Arithmetic and Logic Operations, Fetching a word from Memory, Storing a word in Memory - Execution of a Complete Instruction, Branch Instructions - Hardwired Control - Complete Processor - Microprogrammed Control - Pipelining ..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding Microprocessor.. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Microprocessor.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline architecture of a Pentium Processor. (3 marks)

Answer: Begin with the standard definition or identity of *Outline architecture of a Pentium Processor*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website

Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the features of Multicore Processors 3 Understanding Basic x86 Architecture - Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family - Features of 8086 -Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – General Architecture of a Pentium Processor - Multi core Processor Text / Reference: Author /Book T/R Title T1 Carl Hamachar, Computer Organization, McGraw Hill, 5th ed. Lyla B Das, The x86 Microprocessors- Architecture, Programming and T2 Interfacing, Pearson William Stallings, Computer Organization and Architecture, Pearson R1 Education , 8th ed. R2 Morris Mano, Computer System Architecture , Prentice Hall of India- 2002 R3 John Hayes, Computer Architecture and Organization ,McGraw Hill-1998 Sunil Mathur, Microprocessor 8086 Architecture, Programming and R4 Interfacing, PHI,2011 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106108100/ 2 www.tutorialsspace.com/Computer-Architecture-And-Organization*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding Computer system. Illustrate the basic operational concepts of a**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding mapped I/O and Program controlled I/O. Illustrate the working of interrupts in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding processor**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Summarize the history of Microprocessors 1 Understanding Illustrate the architecture 8086/8088**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://nptel.ac.in/courses/106108100/

- www.tutorialsspace.com/Computer-Architecture-And-Organization

IN-PAGE SEARCH

Search this lesson